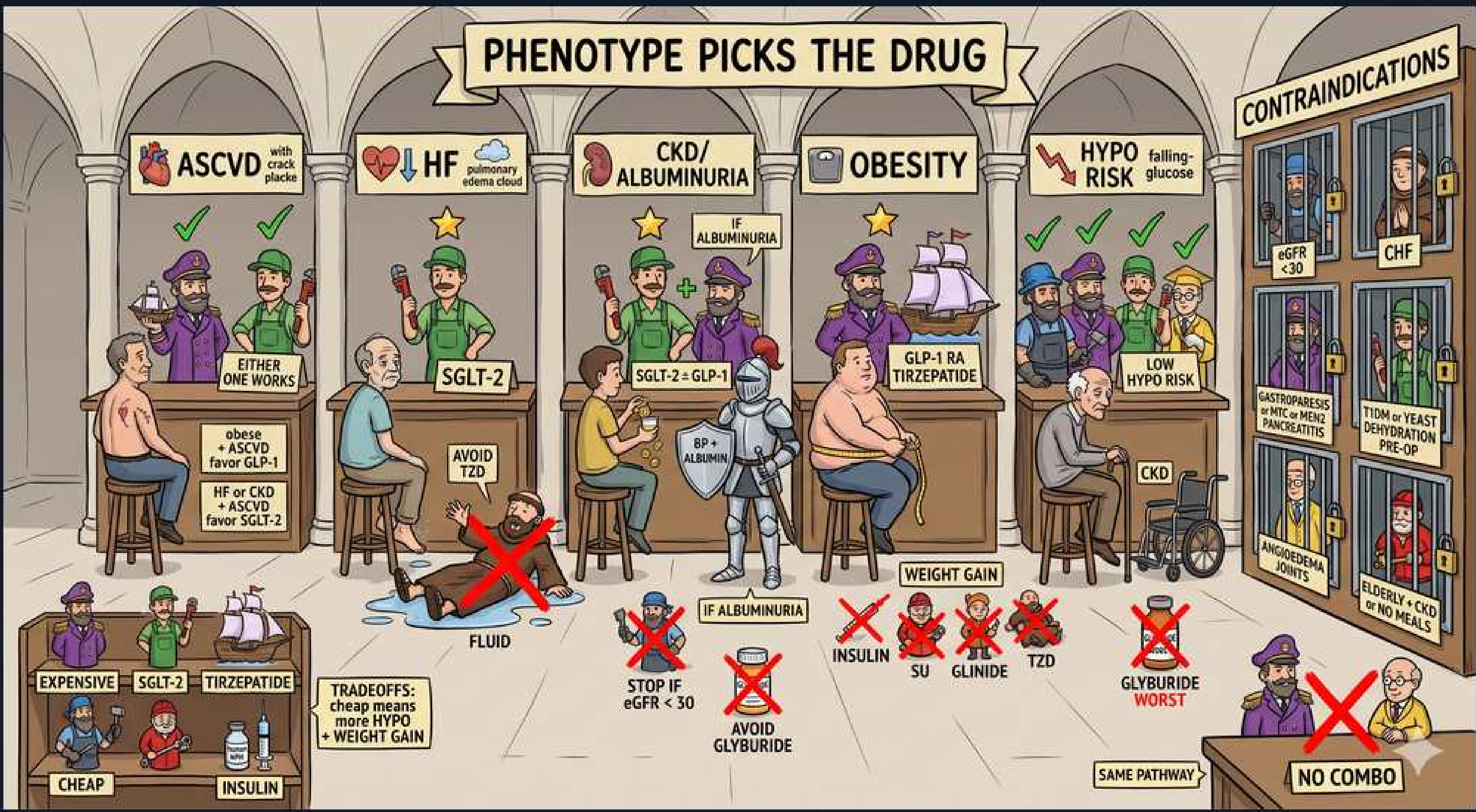
Diabetes — Scene 2: T2DM Phenotype Picker's Pharmacy



1. Phenotype picks the drug

WHAT YOU SEE

Banner: 'Phenotype picks the drug.'

WHAT IT ENCODES

Modern ADA/EASD algorithm: metformin is no longer mandatory first if a compelling comorbidity exists, but in practice it remains the cheap backbone. After (or alongside) metformin, the patient's dominant comorbidity — established ASCVD, heart failure, CKD/albuminuria, obesity, or hypoglycemia risk — selects the second agent INDEPENDENT of how high the A1c is. The overall glycemic target is A1c <7% for most adults (relaxed to <8% in frail/elderly, tightened to <6.5% in young/short-duration if achievable without hypoglycemia). Mnemonic: 'pick the organ you most want to protect, then pick its drug.'

BOARD TRAP

Wrong answer: 'choose the agent that lowers A1c the most.' Discriminator — once a patient has ASCVD/HF/CKD, you choose a drug with proven organ benefit (SGLT2i or GLP-1 RA) regardless of A1c, not the most potent glucose-lowering agent. The boards reward outcome-driven, not glucose-driven, selection.

2. ASCVD counter

WHAT YOU SEE

'ASCVD' (with mark, heart attack): either GLP-1 or SGLT2 works; ASCVD favor GLP-1.

WHAT IT ENCODES

Established atherosclerotic CV disease (prior MI, stroke, PAD, or significant stenosis): both GLP-1 RA and SGLT2i reduce MACE, but GLP-1 RA is favored for atherosclerotic events (LEADER liraglutide, SUSTAIN-6 semaglutide, REWIND dulaglutide all showed MACE reduction). GLP-1 RAs work partly via anti-atherosclerotic and weight/BP effects. SGLT2i (EMPA-REG empagliflozin, CANVAS canagliflozin) also cut CV death, with their edge being heart-failure/renal outcomes. Mnemonic: 'arteries → GLP-1; pump/kidney → SGLT2.'

BOARD TRAP

Wrong answer: 'add a sulfonylurea or DPP-4 inhibitor because A1c is still high.' Discriminator — SU and DPP-4i are CV-NEUTRAL (saxagliptin even raised HF hospitalization in SAVOR-TIMI). In a patient with ASCVD you must pick a class with demonstrated MACE reduction (GLP-1 RA or SGLT2i).

3. HF counter — SGLT2

WHAT YOU SEE

'HF' (pulmonary edema cloud): SGLT2; avoid TZD.

WHAT IT ENCODES

Heart failure (HFrEF AND HFpEF) → SGLT2i is the agent of choice. Empagliflozin (EMPEROR-Reduced/Preserved) and dapagliflozin (DAPA-HF/DELIVER) reduce HF hospitalization and CV death even in NON-diabetics, which is why SGLT2is are now part of the four-pillar HF regimen. GLP-1 RAs do NOT carry this HF benefit. TZDs cause sodium/water retention and worsen HF. Mnemonic: 'failing pump loves a flozin.'

BOARD TRAP

Wrong answer: 'pioglitazone, since it improves insulin sensitivity.' Discriminator — TZDs (rosi- AND pioglitazone) are contraindicated in symptomatic NYHA III–IV HF because PPAR-γ activation causes fluid retention and edema. Never pick a TZD for the HF phenotype; pick an SGLT2i.

4. CKD / albuminuria counter

WHAT YOU SEE

'CKD/Albuminuria': SGLT2 +/- GLP-1; stop if eGFR <30; BP +/- albuminuria.

WHAT IT ENCODES

Diabetic kidney disease, especially with albuminuria → SGLT2i is renoprotective (CREDENCE canagliflozin, DAPA-CKD dapagliflozin slowed progression to ESRD and reduced albuminuria via afferent arteriolar constriction lowering intraglomerular pressure). Can INITIATE SGLT2i down to eGFR ~20–25 for renal protection even though glucose-lowering wanes below ~45; the renal indication differs from the glycemic one. Add GLP-1 RA if more glucose lowering or CV/weight benefit is needed. Mnemonic: 'flozin for the failing filter.'

BOARD TRAP

Wrong answer: 'continue glyburide and titrate up.' Discriminator — glyburide has active renally-cleared metabolites that accumulate in CKD and cause prolonged hypoglycemia; it is the worst SU in renal impairment. Choose SGLT2i ± GLP-1 RA and, if a secretagogue is unavoidable, use glipizide (hepatic metabolism).

5. Obesity counter

WHAT YOU SEE

'Obesity': GLP-1 RA / tirzepatide; weight gain agents crossed out.

WHAT IT ENCODES

Obese phenotype → GLP-1 RA (semaglutide ~10–15% weight loss; liraglutide) or the dual GIP/GLP-1 agonist tirzepatide (SURMOUNT: up to ~20% weight loss, the most potent). These slow gastric emptying and act on hypothalamic satiety. SGLT2i give modest weight loss (~2–3 kg via glucosuria). Mnemonic: 'tirzepatide = two incretins, top weight loss.'

BOARD TRAP

Wrong answer: 'add basal insulin or a sulfonylurea to bring glucose down faster.' Discriminator — insulin, SU, glinides, and TZDs all cause WEIGHT GAIN, the opposite of the goal. In the obese patient pick a GLP-1 RA or tirzepatide; reserve insulin for severe hyperglycemia/catabolism.

6. Hypo-risk counter

WHAT YOU SEE

'Hypo risk' (falling glucose): pick low-hypo agents.

WHAT IT ENCODES

When hypoglycemia is dangerous (elderly, frail, commercial drivers, lives alone, CKD) → choose agents that do NOT cause hypoglycemia as monotherapy: metformin, SGLT2i, GLP-1 RA, DPP-4i, and TZD. These either reduce hepatic glucose, increase glucosuria, or stimulate insulin GLUCOSE-DEPENDENTLY (incretins). Mnemonic: 'incretins are smart — they only push insulin when sugar is high.'

BOARD TRAP

Wrong answer: 'glyburide is fine, it's cheap and effective.' Discriminator — sulfonylureas and glinides close the K-ATP channel and force insulin secretion regardless of glucose, so they (and insulin) carry the HIGHEST hypoglycemia risk. They are the wrong pick for any hypo-vulnerable patient.

7. Contraindications wall

WHAT YOU SEE

'Contraindications': eGFR <30, CHF (TZD), DKA-prone, dehydration pre-op, MTC/MEN2.

WHAT IT ENCODES

Class-specific contraindications: SGLT2i — euglycemic DKA risk (hold for surgery/illness/dehydration, avoid in T1DM, glucose-lowering limited at very low eGFR); TZD — symptomatic CHF, prior bladder cancer concern (pioglitazone), high fracture risk; GLP-1 RA — personal/family history of medullary thyroid carcinoma or MEN2 (rodent C-cell tumors), prior pancreatitis; metformin — eGFR <30, IV contrast, sepsis/hypoperfusion (lactic acidosis). Mnemonic: 'GLP-1 hates the thyroid C-cell; flozin hates dehydration; TZD hates the failing heart.'

BOARD TRAP

Wrong answer: 'start a GLP-1 RA in a patient with a thyroid nodule and family history of medullary thyroid cancer.' Discriminator — personal/family MTC or MEN2 is an absolute contraindication to GLP-1 RAs (and tirzepatide) due to the C-cell tumor signal; choose SGLT2i or another class instead.

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8. Avoid TZD — fluid

WHAT YOU SEE

Figure slipping on fluid: 'avoid TZD — fluid.'

WHAT IT ENCODES

TZDs (pioglitazone, rosiglitazone) activate PPAR-γ, which upregulates renal ENaC sodium reabsorption causing fluid retention, peripheral edema, weight gain, and precipitation/worsening of heart failure. Effect is dose-dependent and worse when combined with insulin. Mnemonic: 'PPAR-γ → puddles (edema).'

BOARD TRAP

Wrong answer: 'TZD edema is just cosmetic, keep the drug and add a thiazide.' Discriminator — TZD fluid retention is mechanistically distal-nephron and resistant to thiazides/loops; the correct action is to STOP the TZD, especially in any patient with HF or volume sensitivity. Do not pick a TZD when fluid status matters.

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9. Stop SGLT2 if eGFR <30 / avoid glyburide

WHAT YOU SEE

'Stop if eGFR <30'; 'avoid glyburide' in CKD.

WHAT IT ENCODES

SGLT2i glucose-lowering efficacy depends on filtered glucose load, so it falls as eGFR declines (minimal glycemic effect below ~45); however renal/CV protection persists and they may be continued/initiated to lower eGFR thresholds for those indications. Glyburide is renally cleared with active metabolites → accumulation and prolonged hypoglycemia in CKD/elderly. Mnemonic: 'flozin = weak sugar drug but strong organ drug at low GFR; glyburide = land mine in CKD.'

BOARD TRAP

Wrong answer: 'expect a big A1c drop from an SGLT2i started at eGFR 25.' Discriminator — at low eGFR the SGLT2i barely lowers glucose (you keep it for kidney/heart protection, not A1c). If you need glycemic control there, lean on insulin or a GLP-1 RA, and never reach for glyburide.

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10. Glyburide — worst

WHAT YOU SEE

Wheelchair figure: 'glyburide — worst.'

WHAT IT ENCODES

Among sulfonylureas, glyburide (glibenclamide) is the worst for hypoglycemia: long-acting, renally cleared, with active metabolites — high risk in the elderly and in CKD (severe, prolonged, sometimes overnight hypoglycemia). If a SU must be used, prefer glipizide (short-acting, hepatic metabolism) or glimepiride. Mnemonic: 'glyBURIDE buries the elderly.'

BOARD TRAP

Wrong answer: 'glyburide for an 80-year-old with CKD because it's inexpensive.' Discriminator — age >65 and CKD both amplify glyburide hypoglycemia; the Beers list flags it. Pick glipizide if a secretagogue is needed, or better, a low-hypo agent (DPP-4i, GLP-1 RA, SGLT2i).

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11. Cheap vs expensive

WHAT YOU SEE

Expensive: SGLT2, tirzepatide; cheap: SU, metformin, insulin (older).

WHAT IT ENCODES

Cost/benefit tradeoff: GLP-1 RAs, SGLT2is, and tirzepatide are expensive but deliver CV/renal/weight benefit with low hypoglycemia; metformin, sulfonylureas, and older human insulins (NPH/regular) are cheap but SU/insulin bring weight gain and hypoglycemia. Mnemonic: 'cheap = secretagogues (gain + hypo); pricey = protectors (lose weight, low hypo).'

BOARD TRAP

Wrong answer: 'cost should never influence the choice.' Discriminator — when there is NO compelling comorbidity and budget is limited, a cheap SU added to metformin is acceptable; but when ASCVD/HF/CKD is present, the outcome benefit of SGLT2i/GLP-1 RA outweighs cost. Tailor to both the phenotype and access.

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12. No combo — same pathway

WHAT YOU SEE

'No combo / same pathway' (DPP-4 + GLP-1 crossed).

WHAT IT ENCODES

Never combine two drugs that hit the same target. DPP-4 inhibitors raise endogenous GLP-1/GIP, so adding them to an injectable GLP-1 RA gives no extra benefit (the GLP-1 RA already supraphysiologically activates the receptor) — redundant and wasteful. Likewise, don't stack two insulin secretagogues (SU + glinide) or a secretagogue + prandial insulin (additive hypoglycemia). Mnemonic: 'one knob per pathway.'

BOARD TRAP

Wrong answer: 'add sitagliptin to a patient already on semaglutide to boost the incretin effect.' Discriminator — DPP-4i + GLP-1 RA is the classic redundant combination; discontinue the DPP-4i when starting a GLP-1 RA. The correct add-on is a different mechanism (metformin, SGLT2i, etc.).
